

Lecture #3

Content to be covered

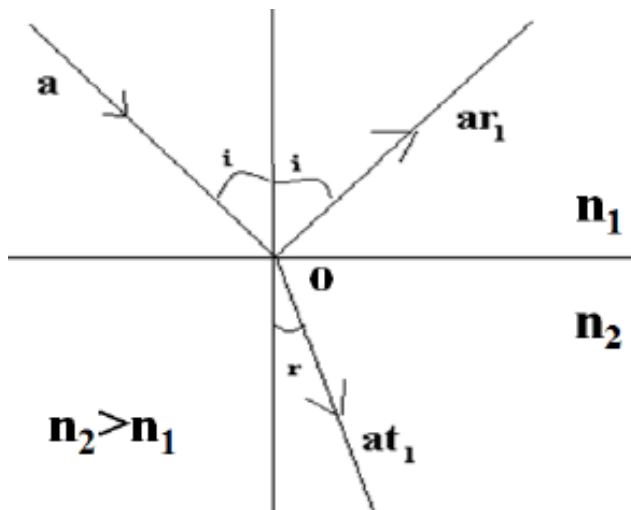
- ❑ Principle of optical reversibility
 - ❖ Phase change at reflection
 - ❖ Stokes' law
- ❑ Interference by division of amplitude
- ❑ Thin films Interference
 - ❖ Path difference (Cosine Law)
 - ❖ Some case study

Resources to be consulted

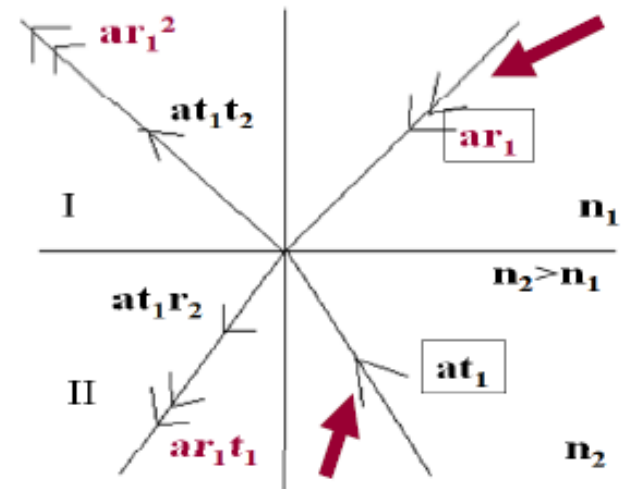
- ❑ OPTICS by Ajay Ghatak

Principle of optical reversibility

In the absence of any absorption, a light ray that is reflected or refracted will retrace its original path if its direction is reversed.



$a \rightarrow$ amplitude of incident ray
 $r_1 \rightarrow$ reflection coefficient
 $t_1 \rightarrow$ transmission coefficient
 $ar_1 \rightarrow$ amplitude of reflected ray
 $at_1 \rightarrow$ amplitude of refracted ray



$r_1, r_2 \rightarrow$ reflection coefficients
 $t_1, t_2 \rightarrow$ transmission coefficients
 $n_1, n_2 \rightarrow$ refractive index of two media

Optical Reversibility

According to principle of optical reversibility the two rays of amplitudes ar_1^2 and at_1t_2 must combine to give the incident ray of figure 1.

$$\text{So, } ar_1^2 + at_1t_2 = a \Rightarrow t_1t_2 = 1 - r_1^2 \quad \dots(1) \quad \text{Stokes' relation}$$

The two rays of amplitudes at_1r_1 and at_1r_2 must cancel each other.

$$\text{So, } at_1r_1 + at_1r_2 = 0 \Rightarrow r_2 = -r_1 \quad \dots(2) \quad \text{Stokes' relation}$$

Phase change π occurs when light gets reflected from denser medium.

no phase change will occur when light gets reflected by a rarer medium.

□Eq.(2) represents that the coefficient of reflection from the surface of a denser medium is equal in magnitude to the coefficient of reflection from the surface of a rarer medium but opposite in sign.

□Intensity of reflected light is the same for a ray incident from either side of the boundary.

□Eq.(2) does not give any information as to which of the two reflection gives the phase change.

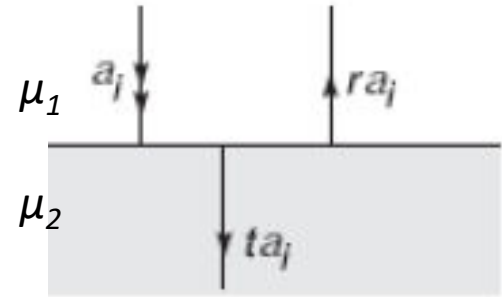
Reflection and Transmission coefficients

Assuming a light beam is incident normally on a interface. Then the amplitudes of the reflected and transmitted beams are related to that of the incident beam through the following relations:

$$a_r = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} a_i$$

$$a_t = \frac{2\mu_1}{\mu_1 + \mu_2} a_i$$

where a_i , a_r , and a_t are the amplitudes of the incident beam, reflected beam, and transmitted beam, respectively.



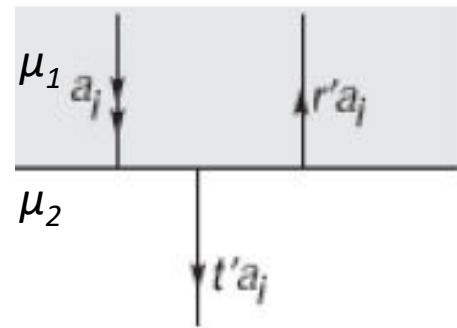
One can observe that when $\mu_2 > \mu_1$ amplitude a_r is negative, showing that when a reflection occurs at a denser medium, a phase change of π occurs. The reflection and transmission coefficients are given by:

$$r = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}$$

$$t = \frac{2\mu_1}{\mu_1 + \mu_2}$$

If r' and t' are the reflection and transmission coefficients when light is incident on a medium of refractive index μ_1 then

$$r' = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} \quad t' = \frac{2\mu_2}{\mu_1 + \mu_2} \Rightarrow 1 - tt' = 1 - \frac{4\mu_1\mu_2}{(\mu_1 + \mu_2)^2} = r^2$$



Interference by Division of Amplitude

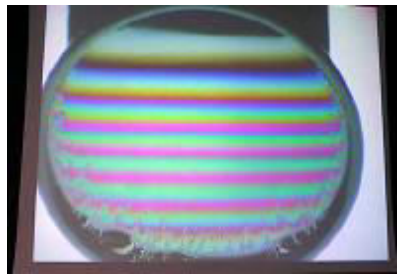
Suppose a light wave is incident on a **half silvered mirror** or simply a sheet of glass, a part of the wave is transmitted and part reflected. Both the transmitted and reflected waves have lower amplitudes than the original one. It can be said figuratively that the amplitude has been split.

Examples

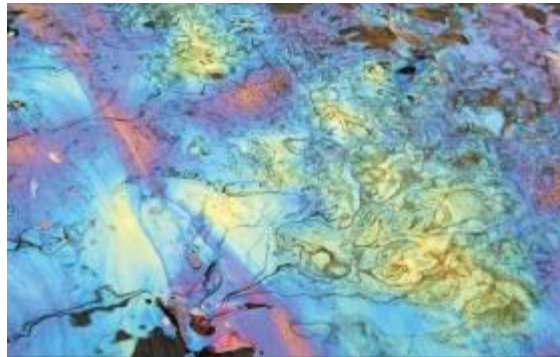
- ❖ Interference by thin films
- ❖ Newton's Ring
- ❖ Michelson's Interferometer
- ❖ Feby Perot's Interferometer



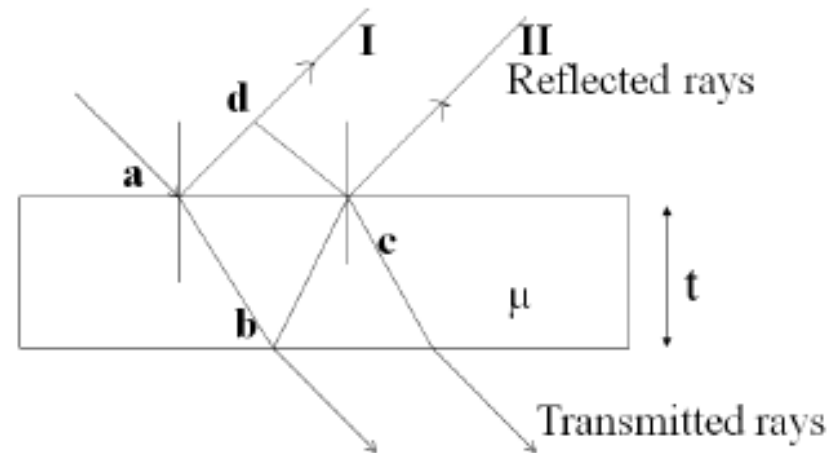
Anti-reflecting thin films



Colours of soap bubble



Colours of oil film on water



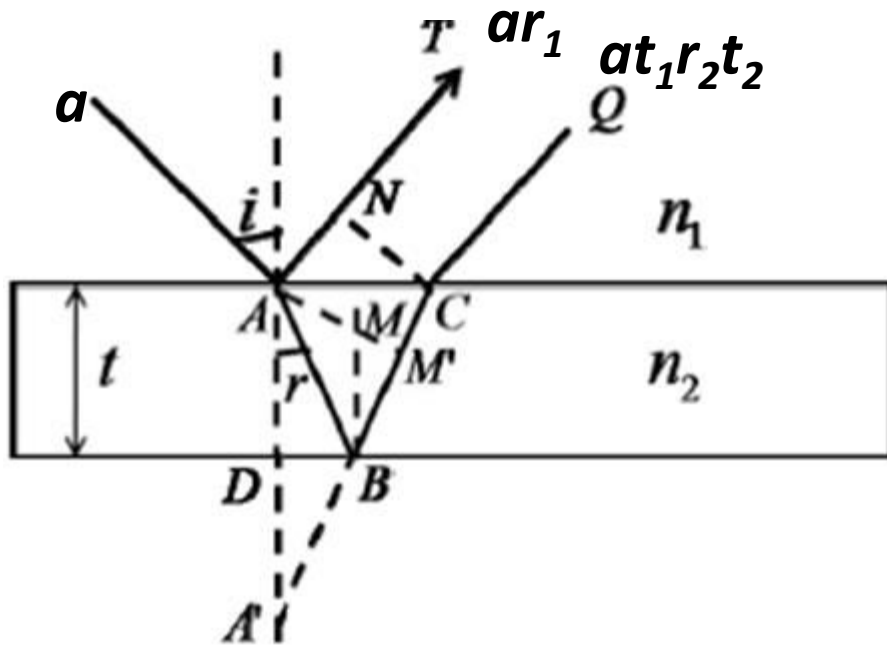
If plane wave falls on a thin film then the wave reflected from the upper surface interferes with the wave reflected from the lower surface.

For thin-film optics:

$$t_{\text{thin-film}} \sim 1 \mu\text{m}$$

($t_{\text{thin-film}} \rightarrow$ order of the λ visible light)

Interference In Thin Films



Thus, under above condition, Interference in the thin film would essentially be two beam interference. But if the reflectivity is large, one obtains a multiple beam interference.

From Stokes relation

$$t_1 t_2 = 1 - r_1^2$$

$$at_1 t_2 r_2 = a(1 - r_1^2) r_2$$

If $|r_1| = |r_2| \ll 1$

$$\Rightarrow at_1 t_2 r_2 = ar_2$$

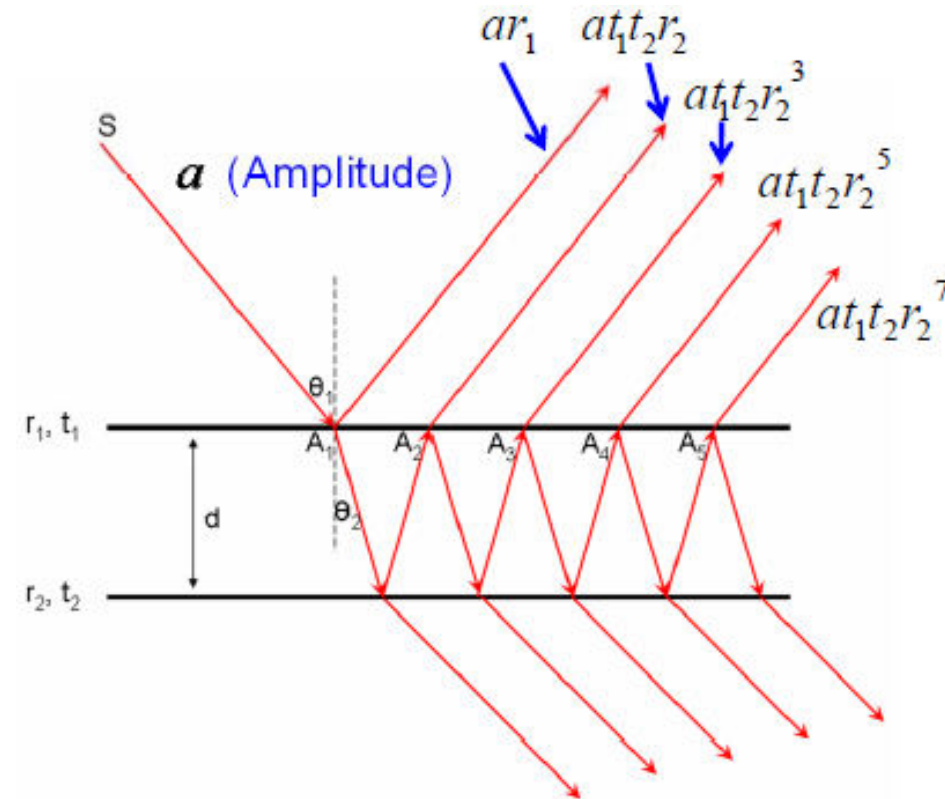
For next wave, amplitude is

$$at_1 t_2 r_2^3 = a(1 - r_1^2) r_2^3$$

$$at_1 t_2 r_2^3 = ar_2^3 \ll 1$$

amplitude of all other waves would be very-very small.

Multiple Beam Interference In Thin Films



$$A = at_1t_2r_2 + at_1t_2r_2^3 + at_1t_2r_2^5 + \dots$$

$$= at_1t_2r_2[1 + r_2^2 + r_2^4 + \dots]$$

$$= at_1t_2r_2 \times \frac{1}{1 - r_2^2}$$

$$= a(1 - r_2^2)r_2 \times \frac{1}{1 - r_2^2}$$

as per principle of reversibility

$$t_1t_2 = 1 - r_1^2 \quad \& \quad r_2 = -r_1$$

$$A = ar_2 = -ar_1$$

So, the resultant amplitude of rays reflected from the bottom surface of thin film (2,3,4,...) is equal in magnitude of the amplitude of ray 1 but out of phase with it.

Interference In Thin Film of Uniform thickness 't' **Reflection**

The optical path difference between AT and CQ

$$\Delta = n_2(AB + BC) - n_1 AN$$

$$\angle DAB = \angle ABM = \angle MBC$$

Now $\angle ABD = \frac{\pi}{2} - r$

and $\angle A'BD = \pi - \left[\left(\frac{\pi}{2} - r \right) + r + r \right]$

$$\Rightarrow \angle A'BD = \frac{\pi}{2} - r$$

Thus $AB = BA'$ and $AD = DA' = t$ Thus $AN = AC \sin i$

or $AB + BC = A'B + BC = A'C$

$$\Delta = n_2(AB + BC) - n_1 AN$$

$$\Rightarrow \Delta = n_2 A'C - n_1 AN$$

$$\Rightarrow AN = \frac{CM'}{\sin r} \sin i = \frac{n_2}{n_1} CM'$$

$$[n_1 \sin i = n_2 \sin r]$$

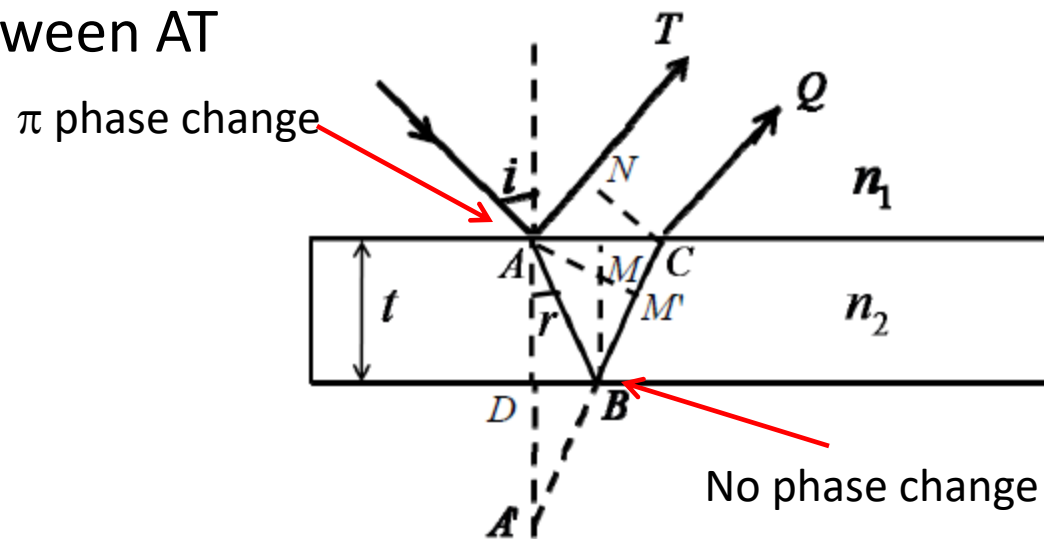
Now $\angle NCA = i$

$$\Delta = n_2 A'C - n_1 AN$$

$$\Rightarrow \Delta = n_2 A'M' = n_2 \times 2t \cos r$$

$$\Rightarrow \Delta = 2n_2 t \cos r$$

COSINE LAW



Interference In Thin Films

Considering a phase change π equivalent to a path difference $\lambda/2$ due to reflection from the surface of optically denser medium.

The net path difference will be $\Delta = 2\mu t \cos r - \frac{\lambda}{2}$

So condition for **maxima** will be

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$\Rightarrow 2\mu t \cos r = (2n + 1)\lambda / 2$$

for **destructive** interference

$$2\mu t \cos r - \frac{\lambda}{2} = (2n + 1)\lambda / 2$$

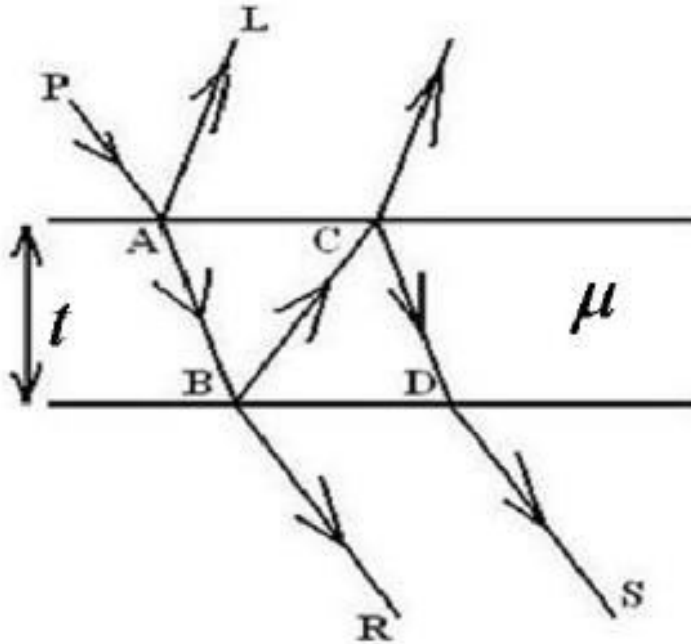
$$\Rightarrow 2\mu t \cos r = (n + 1)\lambda = m\lambda$$

❑ Here, interference pattern will not be perfect because intensities AT and CQ will not be the same and their amplitude are different.

❑ Amplitude depends on amount of light reflected and transmitted through the films.

❑ Intensity never vanishes completely and perfectly dark fringes will not be observed. But, for multiple reflection intensity of minima will be zero.

Interference In Thin Film of Uniform thickness 't' **Transmission**



for **constructive** interference

$$2\mu t \cos r = n\lambda$$

for **destructive** interference

$$2\mu t \cos r = (2n + 1)\lambda / 2$$

$$n = 0, 1, 2, 3, \dots$$

Excessively thin film in air i.e. $t \ll \lambda$

General conditions for constructive and destructive interference are, respectively,

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$$

But under the given condition i.e. $t \ll \lambda$ $\Delta \approx \lambda / 2$ **Thus film will appear dark**

Excessively thick film in air i.e. $t \gg \lambda$

$$\Delta = 2\mu t \cos r$$

Thus conditions for constructive and destructive interference would be, respectively,

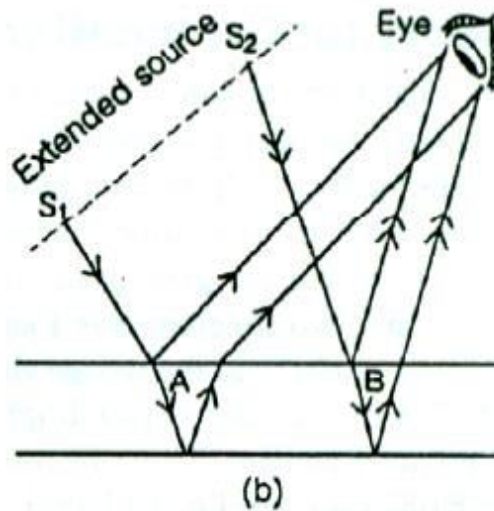
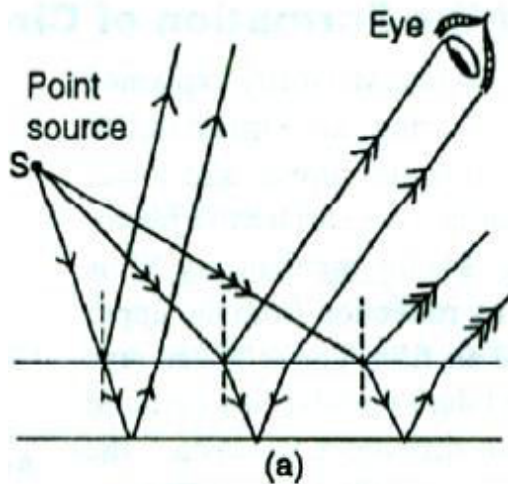
$$\Delta = 2\mu t \cos r = n\lambda$$

$$\Delta = 2\mu t \cos r = (2n+1)\frac{\lambda}{2}$$

Need of Extended source

❑ When a thin film is illuminated by a point source then, for different incident rays, different pairs of interfering rays are obtained along widely different angles. All these pairs can't be received by the eye. The rays only from a small portion of the film can enter the eye. Hence the entire film can't be seen by the eye placed in a fixed position.

❑ When the film is illuminated by an extended source, the rays from different points of the source are reflected from different parts of the film so as to enter the eye placed in a fixed position. Hence one can see the entire film simultaneously.



An extended source is necessary to enable the eye to see a large area of the film simultaneously.